PROJECT ANALYSIS

Macchanger and Wazuh Open Source SIEM

Student: Shaik Subahan
Project Team ID: PTID-CSPP-AUG-26-540

1. Problem and Purpose

The project provides hands-on exposure to two cybersecurity concepts: MAC address randomization at the network-interface level and security monitoring using an open-source SIEM platform. The practical was designed to build familiarity with Linux commands and verify observable system behavior in a controlled virtual lab.

2. Method

The work was carried out in Kali Linux inside VMware Workstation. For the Macchanger task, the current and permanent MAC addresses were checked, the interface was temporarily disabled, a random MAC address was generated, and the interface was re-enabled. The changed address was verified and the permanent MAC was restored. For Wazuh, the Manager service was started and checked using systemctl. Listening ports were inspected using ss and filtered for ports 1514, 1515, and 55000.

3. Findings

• Macchanger generated a randomized MAC address that differed from the permanent VMware address.
• The permanent MAC was restored and the final check showed the current and permanent values matching.
• Wazuh Manager was reported as active (running).
• Ports 1514, 1515, and 55000 were observed listening.
• The local Wazuh API responded, but authentication returned Unauthorized/Invalid credentials; successful API authentication is not claimed.
• The ping test received 2 replies from 4 packets, indicating 50% packet loss during that test.

4. Analysis

The Macchanger output provides direct evidence of address randomization and restoration. The Wazuh service status and listening sockets provide service-level evidence that the Manager and associated listeners were active. These checks do not by themselves demonstrate alert generation, agent enrollment, or a full Wazuh dashboard workflow. Those activities were outside the completed scope of this constrained lab.

5. Conclusion

The practical met its focused objectives: demonstrate MAC address randomization and restoration, and verify the Wazuh Manager service and relevant listening ports. The main learning outcomes were Linux network-interface administration, service verification, and introductory understanding of SIEM components and their roles.

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